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Module 1 Lab 3

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In Chapter 1.9.4 Shah describes Data Analytics to being prone to bias. When a dataset is analyzed to get information, that information is a direct result of the dataset used. "The output of the data analytics is limited to the details of the input." This is because, if a dataset is skewed to a particular outcome, it's not the most accurate representation of the whole, so it could be a mistake to say that the best average is the best choice. The max rating is not the best way to see which company provides the best service, and the min rating tells about the most unhappy customer. The risk involved is that a very low rating can pull down the average, and this could be a one time occurrence. This is why variance helps to tell if the average is a dependable method to choose the best company. If the existing data structure contains bias, by establishing the bias, you can prevent it's amplification.

Nathan Yau of Flowing Data says three basic skills of Data Science are: **1. A strong knowledge of statistics**, (avoid misinterpreting a correlation, or extrapolating too much from a small sample size) **2. Analyze a dataset with a programming language.** **3. Help someone less experienced to visualize the data.**

```
In [1]: import pandas as pd

# our dictionary
data = { 'Geico': [4.7, 8.3, 9.2], 'Progressive':[7.4, 6.7, 8.1], 'USAA': [3.8, 6.3, 8.1] }
df = pd.DataFrame(data)
print(df)
```

	Geico	Progressive	USAA
0	4.7	7.4	3.8
1	8.3	6.7	6.3
2	9.2	8.9	8.1

Pandas lets us visualize our data above.

```
In [2]: geico_ratings = [4.7, 8.3, 9.2]
progressive_ratings = [7.4, 6.7, 8.9]
usaa_ratings = [3.8, 6.3, 8.1]
```

These three lists above can be used with Numpy to find the **max, min, mean, median, variance, and standard deviation**. The next 3 code blocks will print these statistics.

```
In [3]: import numpy as np
geico_max = np.max(geico_ratings)
geico_min = np.min(geico_ratings)
geico_mean= np.mean(geico_ratings)
geico_variance = np.var(geico_ratings)
geico_std = np.std(geico_ratings)
geico_median = np.median(geico_ratings)

print("geico max",geico_max)
print("geico min",geico_min)
print("geico mean {:.4f}".format( geico_mean))
print("geico variance {:.4f}".format( geico_variance))
print("geico std {:.4f}".format(geico_std))
print("geico median {:.4f}".format(geico_median))
```

```
geico max 9.2
geico min 4.7
geico mean 7.400000
geico variance 3.780000
geico std 1.944222
geico median 8.300000
```

```
In [4]: progressive_max = np.max(progressive_ratings)
progressive_min = np.min(progressive_ratings)
progressive_mean= np.mean(progressive_ratings)
progressive_variance = np.var(progressive_ratings)
progressive_standarddev = np.std(progressive_ratings)
progressive_median = np.median(progressive_ratings)
print("progressive max",progressive_max)
print("progressive min",progressive_min)
print("progressive mean {:.4f}".format( progressive_mean))
print("progressive variance {:.4f}".format( progressive_var))
print("progressive std {:.4f}".format(progressive_standarddev))
print("progressive median {:.4f}".format(progressive_median))
```

progressive max 8.9
progressive min 6.7
progressive mean 7.666667
progressive variance 0.842222
progressive std 0.917727
progressive median 7.400000

```
In [5]: usaa_max = np.max(usaa_ratings)
usaa_min = np.min(usaa_ratings)
usaa_mean= np.mean(usaa_ratings)
usaa_variance = np.var(usaa_ratings)
usaa_standarddev = np.std(usaa_ratings)
usaa_median = np.median(usaa_ratings)
print("usaa max",usaa_max)
print("usaa min",usaa_min)
print("usaa mean {:.4f}".format( usaa_mean))
print("usaa variance {:.4f}".format( usaa_variance))
print("usaa std {:.4f}".format(usaa_standarddev))
print("usaa median {:.4f}".format(usaa_median))
```

usaa max 8.1
usaa min 3.8
usaa mean 6.066667
usaa variance 3.108889
usaa std 1.763204
usaa median 6.300000

We can print the mean of each output and take the largest as the best.

```
In [6]: insurance_mean =[geico_mean, progressive_mean, usaa_mean]
insurance_mean_key = {'Geico':[geico_mean], 'Progressive':[
print("The Highest Average (mean):",np.max(insurance_mean))
```

```
print("-"*40)
df = pd.DataFrame(insurance_mean_key)
print(df)
```

The Highest Average (mean): 7.666666666666667

```
-----
      Geico  Progressive      USAA
0      7.4      7.666667  6.066667
```

The Progressive mean is the highest average of each company.

We can print the largest median of each output and take the largest as the best

```
In [7]: insurance_median = [geico_median, progressive_median, usaa_median]
insurance_median_key = {'Geico': [geico_median], 'Progressive': [progressive_median], 'USAA': [usaa_median]}
print("The Highest Median:", np.max(insurance_median))

print("-"*40)
df_2 = pd.DataFrame(insurance_median_key)
print(df_2)
```

The Highest Median: 8.3

```
-----
      Geico  Progressive  USAA
0      8.3      7.4      6.3
```

Geico has the highest median.

So which is better, Progressive or Geico? To address fairness and address biases, we make sure that the data has been cleaned, Each company was allowed only three data points, which is a clean dataset. A data scientist may say that data science is about working with increasingly large datasets in the billions, and this allows models to generalize better. For this example we can look at the standard deviation and the variance between these

three points to choose a company. The one with the lowest variance or spread will tell us there is less fluctuations of service and will be the best. Wikipedia says "Standard deviation is found by taking the square root of the average of the squared deviations of the values subtracted from their average value." By comparing a low variance with a high mean we could decide which company will be best.

```
In [8]: #Standard deviation: find the average, subtract the average
insurance_variances = [geico_variance, progressive_variance]
insurance_standarddev = [geico_standarddev, progressive_standarddev]

min_variance = np.min(insurance_variances)
min_standarddev = np.min(insurance_standarddev)

print("Lowest Variance: ",min_variance)
print("Lowest Standard Deviation: ",min_standarddev)

insurance_variance_key = {'Geico':[geico_variance], 'Progressive':[progressive_variance]}
print("-"*40)
insurance_standarddev_key = {'Geico':[geico_standarddev], 'Progressive':[progressive_standarddev]}

df_3 = pd.DataFrame(insurance_variance_key)
print("[Variance]")
print(df_3)
print("-"*40)
df_4 = pd.DataFrame(insurance_standarddev_key)
print("[Standard Deviation]")
print(df_4)
```

Lowest Variance: 0.8422222222222224

Lowest Standard Deviation: 0.9177266598624138

[Variance]

	Geico	Progressive	USAA
0	3.78	0.842222	3.108889

[Standard Deviation]

	Geico	Progressive	USAA
0	1.944222	0.917727	1.763204

Progressive has the lowest variance, and standard deviation and the highest mean. However Geico has the highest median because 9.2 and 8.3 is higher than Progressive's 8.9 and 7.4 ratings. The minimum rating is engaged, not ignored. To prevent prediction loss and miscalculation we consider the mean, std and median. Although Geico has the highest rating of 9.2, their standard deviation and variance is the highest at 3.7 and 1.9. This makes **Progressive** the best company, with their 8.9 trailing closely to Geico's 9.2 rating.

```
In [9]: # We can add Lists together to make one List.
max_rating = np.max(geico_ratings + progressive_ratings +
print(data)
print("The highest of all ratings:", max_rating)
```

```
{'Geico': [4.7, 8.3, 9.2], 'Progressive': [7.4, 6.7, 8.9],
'USAA': [3.8, 6.3, 8.1]}
```

The highest of all ratings: 9.2

The lowest of all ratings is 3.8 by USAA, this can significantly pull down the company's mean. USAA's high score of 8.1 is pulled down because it's a negative outlier. Negative outliers can affect great ratings, even if the company had the highest rating, the outlier affects the average. When the ratings spread is large it reveals that the company's performance is unstable. So we say that for stability and low variance Progressive is the best company.

```
In [10]: # We can add Lists together to make one List.
min_rating = np.min(geico_ratings + progressive_ratings +
```

```
print(data)
print("The lowest of all ratings:", min_rating)
```

```
{'Geico': [4.7, 8.3, 9.2], 'Progressive': [7.4, 6.7, 8.9],  
'USAA': [3.8, 6.3, 8.1]}
```

```
The lowest of all ratings: 3.8
```

Behzad Ghabaei - End Of Lab

In []: